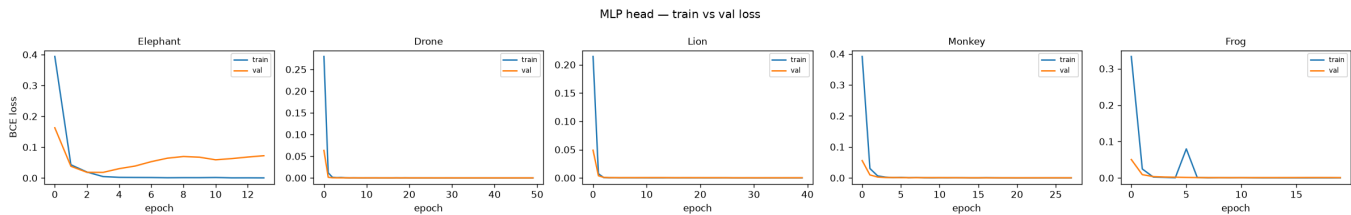


L4 — Small MLP Head

plan.md leg: "Small MLP Head". **Goal:** does non-linearity beat logistic regression? **Model:** YAMNet (frozen) → mean-pool → MLP(1024→256→128→1, sigmoid), Adam lr=1e-3, BCE, 50 epochs, early stopping on val loss (patience=10). **Script:** `experiments/scripts/leg4_mlp_head.py`. **Artifacts:** `figures/mlp_loss_curves.png`, `figures/roc_mlp_head.png`, `experiments/outputs/leg4_mlp_head.json`.

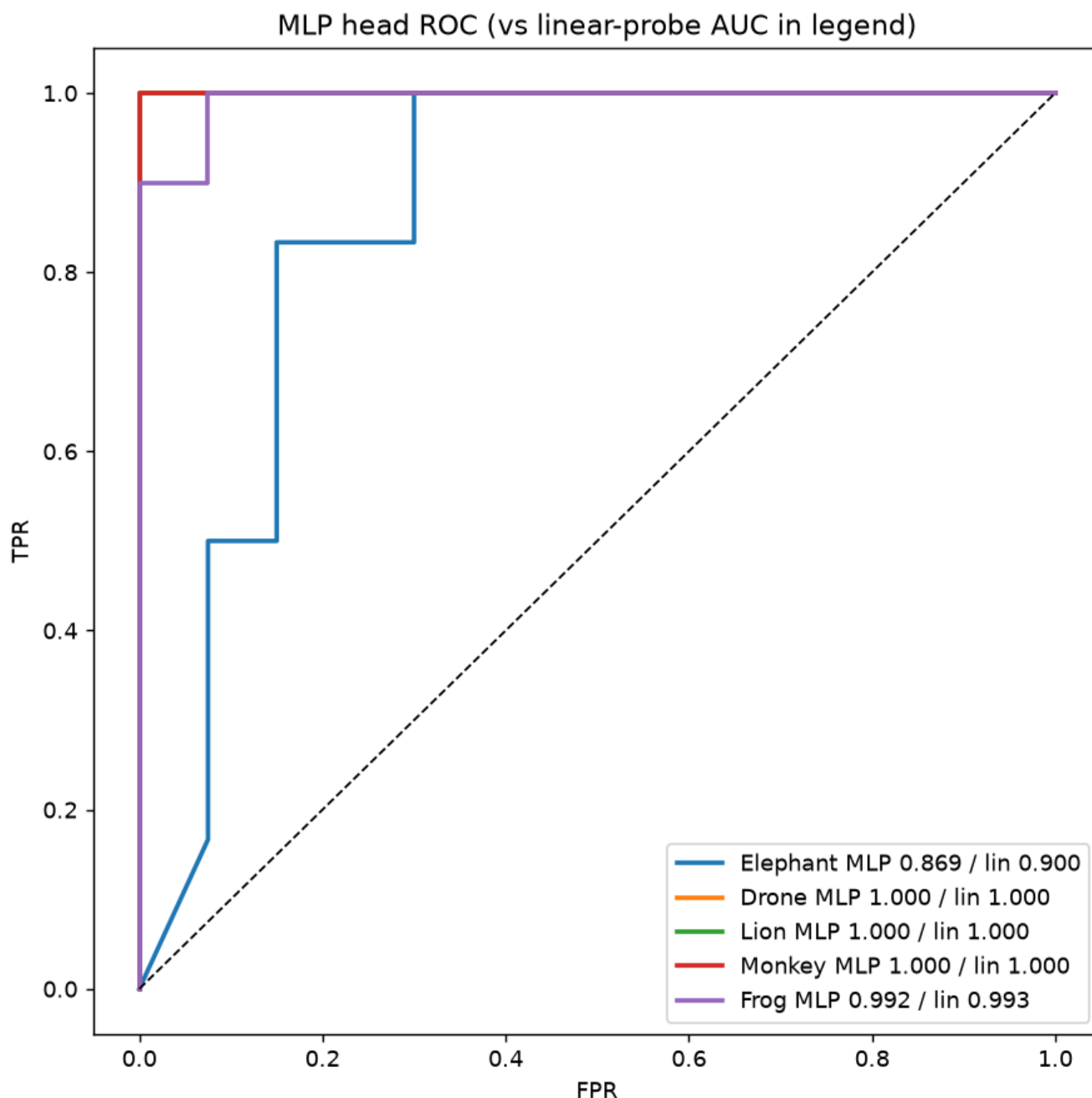
Finding: the MLP does **not** beat the linear probe

class	MLP AUC	linear AUC	Δ	improves >2%?
Elephant	0.869	0.900	-0.031	no
Drone	1.000	1.000	0.000	no
Lion	1.000	1.000	0.000	no
Monkey	1.000	1.000	0.000	no
Frog	0.992	0.993	-0.001	no



The loss curves explain why:

- **Drone / Lion / Monkey / Frog** — train and val loss collapse to ~0 within a few epochs; the problem is already solved linearly, so the MLP just memorizes the same boundary.
- **Elephant** — **textbook overfitting**: train loss → 0 while **val loss rises after ~epoch 3**. Early stopping restored best weights, but the MLP still ends *below* the linear probe (0.869 vs 0.900). With ~40 train clips and an entangled class, the extra capacity hurts.



Interpretation (per plan.md)

"If AUC doesn't improve by >2%: the gain isn't worth the complexity — the problem is the features, not the head."

Confirmed. Non-linearity adds nothing because frozen YAMNet embeddings are already linearly separable for these classes. **Stick with the linear probe** as the lightweight baseline; spend effort on *features/data* (post-ODAS realism, hard negatives, more Elephant separation) — which is precisely what [experiments.pdf](#) Phase 1–2 do — not on a fancier head.

Pass criterion (AUC improves >2%): ❌ for all classes — and that's the informative result, not a failure.